

LONG BEACH, CA, USA

WMO: 722970

Lat: 33.812N

Lon: 118.146W

Elev: 31

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23129

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	41.7	43.9	12.0	10.2	69.3	18.2	13.8	66.1	20.2	56.8	17.7	59.5	1.7	300	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.9	91.8	66.0	88.2	66.0	85.1	65.4	72.1	83.0	70.7	81.1	69.4	79.3	8.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	105.8	76.0	67.2	100.3	75.3	65.9	95.5	74.3	35.7	82.9	34.6	80.9	33.5	79.6	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	13.7	12.1	37.2	100.6	2.0	3.9	35.7	103.4	34.5	105.7	33.4	107.9	31.9	110.7		
(5)			32.7	74.2	2.3	2.4	31.1	75.9	29.7	77.3	28.5	78.6	26.8	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	57.6	57.6	60.4	62.6	65.4	68.8	73.0	74.3	73.0	68.2	62.2	56.9	(6)
(7)		DBStd	7.63	5.08	4.84	4.58	4.88	4.03	3.60	3.62	4.01	4.98	5.05	5.15	4.31	(7)
(8)		HDD50	8	4	2	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD65	1139	235	214	157	102	39	5	0	0	0	19	113	255	(9)
(10)		CDD50	5504	240	214	323	379	477	564	714	755	691	566	366	215	(10)
(11)		CDD65	1159	6	6	14	31	52	119	249	290	242	119	29	2	(11)
(12)		CDH74	6898	131	113	186	312	299	426	1158	1615	1442	839	320	57	(12)
(13)		CDH80	2237	25	31	54	106	104	105	297	526	535	338	109	7	(13)
(14)	Wind	WSAvg	5.1	4.0	4.8	5.4	6.1	6.0	5.9	5.8	5.5	5.1	4.5	3.9	3.9	(14)
(15)	Precipitation	PrecAvg	12.4	2.8	3.2	1.8	0.7	0.3	0.1	0.0	0.1	0.5	0.9	2.1	(15)	
(16)		PrecMax	24.9	12.8	13.0	6.0	2.0	2.1	0.9	0.4	0.1	1.6	4.4	2.2	9.8	(16)
(17)		PrecMin	4.8	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.0	3.2	3.0	1.7	0.6	0.5	0.2	0.1	0.0	0.3	0.9	0.7	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.4	85.3	86.7	90.2	90.3	89.5	90.7	94.2	97.0	96.4	90.0	80.0	(19)
(20)			MCWB	56.6	57.4	59.1	59.7	60.1	67.2	69.1	69.8	69.6	62.6	60.0	54.2	(20)
(21)		2%	DB	77.7	76.9	79.1	82.3	81.5	82.3	86.3	89.1	90.5	88.9	82.6	74.4	(21)
(22)			MCWB	54.7	55.7	56.7	58.6	62.7	66.3	68.5	69.0	68.8	61.8	56.8	54.2	(22)
(23)		5%	DB	72.6	71.0	73.3	76.5	76.0	78.4	83.2	85.6	86.0	82.7	76.6	69.8	(23)
(24)			MCWB	54.1	54.7	56.5	58.2	62.4	65.4	68.0	68.6	67.8	61.6	56.5	53.1	(24)
(25)		10%	DB	67.4	66.3	68.9	71.4	72.1	75.4	80.2	82.4	81.9	77.6	71.9	65.8	(25)
(26)			MCWB	53.7	54.7	56.7	57.8	61.3	64.4	67.6	68.4	67.2	62.4	57.1	53.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	61.9	64.3	65.5	68.5	69.5	72.6	74.0	74.8	70.5	65.1	61.8	(27)
(28)			MCD B	68.1	72.2	75.7	77.7	78.2	82.2	83.9	86.9	85.9	79.3	74.0	67.2	(28)
(29)		2%	WB	59.8	59.8	61.7	62.4	65.6	67.9	70.9	71.9	72.1	68.2	62.8	59.8	(29)
(30)			MCD B	65.2	67.7	70.6	73.5	75.0	79.3	81.1	83.0	82.8	78.4	70.3	65.3	(30)
(31)		5%	WB	58.4	58.4	59.9	60.6	63.8	66.4	69.6	70.6	70.3	66.2	61.3	58.3	(31)
(32)			MCD B	64.3	65.2	67.7	70.1	72.4	76.2	79.2	80.8	81.0	76.1	69.2	63.4	(32)
(33)		10%	WB	57.1	57.1	58.7	59.3	62.3	65.2	68.6	69.3	68.9	64.5	59.9	56.9	(33)
(34)			MCD B	63.5	63.3	65.6	67.7	70.2	73.8	77.6	78.9	78.9	72.9	67.4	61.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.1	17.1	16.5	16.7	14.0	13.9	15.5	16.9	17.1	17.5	19.0	18.7	(35)
(36)			MCDBR	28.7	26.5	26.3	26.8	21.9	19.8	20.0	21.9	23.8	28.0	28.2	26.7	(36)
(37)			MCWBR	12.6	11.7	10.6	10.0	8.1	7.3	7.1	7.6	7.8	10.2	11.6	12.9	(37)
(38)		5% WB	MCD BR	17.8	18.1	18.0	20.1	18.1	18.0	17.6	19.2	20.3	20.1	19.4	16.8	(38)
(39)			MCWBR	9.8	9.5	8.8	8.9	7.5	7.1	6.6	7.2	7.3	8.5	9.8	10.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.352	0.377	0.375	0.376	0.392	0.392	0.376	0.362	0.370	0.321	0.321	(40)	
(41)		taud	2.494	2.469	2.369	2.389	2.417	2.372	2.440	2.470	2.501	2.450	2.588	2.587	(41)	
(42)		Ebn at Noon	276	285	285	290	289	282	282	285	286	274	282	278	(42)	
(43)		Edh at Noon	27	31	37	38	37	39	36	35	32	31	25	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1007	1289	1721	2100	2253	2345	2346	2235	1889	1436	1116	912	(44)	
(45)		RadStd	71	125	95	108	156	146	129	74	81	108	69	69	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	-4.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		+1.12	N/A	-2.97	+1.26	N/A	N/A	N/A	-202	+396	+193

Nomenclature: See separate page